Joint ML Acceleration of the 2D SF_6/Ar ICP Solver: Legacy and LXCat Pipeline Results from Phase-1 6b

Cross-Backend Reproducibility Across Apple MPS and NCSA Delta

Z. Ngan* and M. Abdelghany†

May 2026

Abstract

We report a joint machine-learning acceleration pipeline for the 2D axisymmetric SF_6/Ar inductively coupled plasma (ICP) fluid solver, trained on the Phase-1 6b solver workspace (BC-map-corrected, post-γ_{Al} hold-out refit). The pipeline is run twice in full—once with the Maxwellian-Arrhenius electron-impact rates (legacy) on Apple MPS, and once with the Boltzmann-PINN/Biagi-v10.6 rates (LXCat) on NCSA Delta H100 GPUs—to disentangle backend effects from physics effects. The *same* architecture wins both sweeps (`E3_separate_heads`, shared trunk plus per-species heads), the same training-recipe ablation orders the same way (output-bias initialisation dominates, physics regularisation alone is counter-productive), and the production ensembles converge to nF RMSE values that differ by a factor of ∼1.18 across backends (legacy 0.00448 on MPS; LXCat 0.00381 on Delta). The architectural and qualitative findings replicate exactly across hardware; the absolute scaling does not. The 5-model production ensembles deliver 531× and 874× speedups over the legacy and LXCat solvers respectively, converting several previously expensive workflows (dense parameter sweeps, inverse-design optimisation, Monte-Carlo uncertainty propagation) into interactive ones.

*Apple MPS legacy half: surrogate training and ablation. `ngankait@gmail.com`.
†NCSA Delta LXCat half: cluster runbook and Delta/CUDA training.

Contents

1 Introduction **3**
- 1.1 Scope of this report . . . 3
- 1.2 Motivation . . . 3

2 Phase-1 6b Solver Recap **3**

3 Common ML Pipeline **4**

4 Legacy Half: Maxwellian Arrhenius Rates (Apple MPS) **5**
- 4.1 Hardware and dataset . . . 5
- 4.2 Architecture sweep . . . 5
- 4.3 Ablation study . . . 5
- 4.4 Production ensemble . . . 6

5 LXCat Half: Boltzmann PINN Rates (NCSA Delta) **6**
- 5.1 Hardware and dataset . . . 6
- 5.2 Cluster-portability patches . . . 6
- 5.3 Architecture sweep . . . 6
- 5.4 Ablation study . . . 7
- 5.5 Production ensemble . . . 7

6 Cross-Backend Comparison and Reproducibility **7**
- 6.1 Side-by-side metrics . . . 7
- 6.2 Why the LXCat ensemble is tighter . . . 8

7 Speedup **8**

8 Limitations **9**

9 Future Work **10**

10 Conclusion **10**

1 Introduction

1.1 Scope of this report

This report documents the machine-learning surrogate work performed on the Phase-1 6b solver workspace since 2026-04-22. It supplements (it does not replace) the existing standalone surrogate report at `Plasma Chemistry Module/SF6_surrogate_and_LXCat/report/main.tex`, which presents the original v1 numbers from a single-backend (Mac/MPS) training run. The contribution of *this* report is to present the results of the two parallel training runs that this team carried out in late April / early May 2026:

1. A **legacy** run (Maxwellian-Arrhenius rates) on the author's Apple MPS hardware—this side was performed by Z. Ngan and is committed in `8837c94`.
2. An **LXCat** run (Biagi/Boltzmann-PINN rates) on the NCSA Delta cluster (H100 / A100 nodes)—this side was performed by M. Abdelghany and is committed in `dc52a5f`, `81d6531`.

Both runs use the *same* dataset grid, the *same* candidate architectures, and the *same* training recipe; the only intentional differences are (i) the electron-impact rate-coefficient source (legacy vs. LXCat) and (ii) the compute backend (Mac/MPS vs. Delta/CUDA).

1.2 Motivation

The Phase-1 6b solver, although accurate enough to reproduce qualitative trends, costs 7.76 s per operating-point evaluation in legacy mode and 14.4 s in LXCat mode. This makes operating-envelope sweeps on the order of 10^3–10^4 points prohibitive; closed-loop recipe optimisation is out of reach. An ML surrogate that reproduces the solver's $\log_{10} n_{\mathrm{F}}$ and $\log_{10} n_{\mathrm{SF_6}}$ outputs to within the solver's own modelling uncertainty would unlock these workflows.

A separate hold-out refit of the wall-recombination coefficient γ_{Al} on the corrected BC map (the *D6 refit*, documented in `docs/notes/DIAGNOSTIC_D6_GAMMA_AL_MEMO.md`) **failed**: no scalar γ_{Al} satisfies both 30 % and 90 % SF_6 compositions simultaneously, with residuals changing sign across the composition axis. This means the residual ∼36 % disagreement between the solver and Mettler's measurements is structural rather than a single-knob calibration error, and accelerating the operating-envelope sweep—rather than retuning γ_{Al}—is the unblock toward Phase-3 multi-parameter calibration. The work in this report is therefore the natural next step after Phase-1 6b.

2 Phase-1 6b Solver Recap

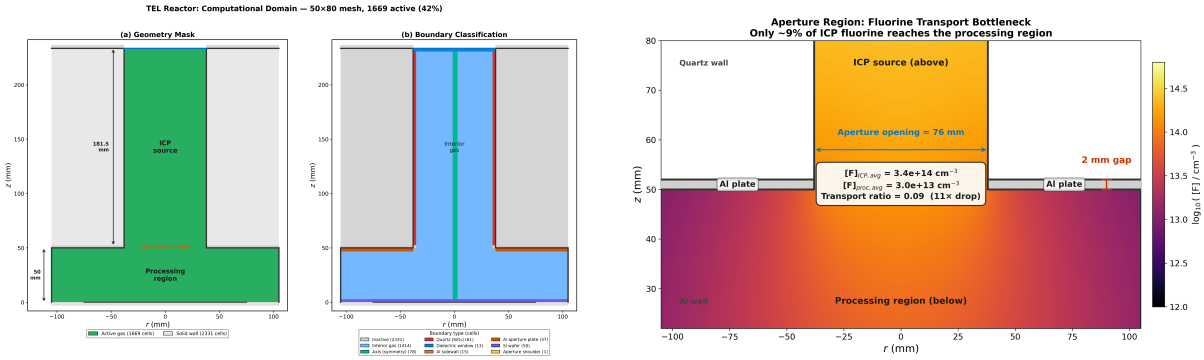
The TEL ICP etcher is modelled as a T-shaped axisymmetric domain: an ICP source region ($R = 38$ mm, $L = 181.5$ mm), a 2 mm aperture, and a processing chamber ($R = 105$ mm, $L = 50$ mm) holding the wafer. The solver discretises the species-continuity, drift–diffusion, and electron-energy equations on a structured (r, z) mesh with Robin boundary conditions at material walls (Kokkoris et al., 2008) and the 54-reaction Lallement (Lallement et al., 2009) chemistry set. Two Phase-1 6b changes since the original main.pdf v1 are material to the data presented here:

BC-map fix. The bottom-boundary mask in `build_geometry_mask` previously labelled the entire $j = 0$ row as `BC_WAFER` (silicon). The 6b correction splits the row at $r = 0.075$ m (the wafer radius) into `BC_WAFER` ($r \leq 0.075$) and `BC_AL_TOP` ($r > 0.075$, the aluminium pedestal-top / chamber-floor annulus). The undercounted aluminium F-sink had area $\sim 0.017\,\mathrm{m}^2$ (∼96 % of the wafer area), so wall-recombination losses on the corrected map are substantially higher. All $[\mathrm{F}]_{\mathrm{c}}$ values shift downward by 4–6 % across the operating envelope as a result.

γ_{Al} hold-out refit (D6). A 9-point sweep of γ_{Al} was performed on the corrected BC map. Fitting on 90 % SF_6 bias-on ($\gamma^*_{\mathrm{Al}} = 0.020$) the fit-point residual was driven to zero by construction, but the held-out conditions failed:

- 90 % SF_6 bias-off: residual −21.7 %
- 30 % SF_6 bias-on: residual +19.8 %
- 30 % SF_6 bias-off: residual +30.0 %

The residual changes sign across composition, ruling out a scalar γ_{Al} as the closure lever. Composition-dependent $\gamma_{\mathrm{Al}}(x_{\mathrm{Ar}})$ is Phase-3 scope. Both this report's surrogates therefore inherit the 6b solver's composition-insensitive ∼66 % wafer F-drop, which disagrees with Mettler's composition-dependent 67–75 % range.



(a) Full T-shaped reactor mask.

(b) Aperture region detail.

Figure 1: 2D axisymmetric reactor geometry used for both halves of the pipeline (figures from (Ngan, 2026)).

3 Common ML Pipeline

Both halves use an identical four-stage pipeline:

1. **Dataset generation**: 220 operating points, $P_{\mathrm{rf}} \in \{200, 300, \ldots, 1200\}$ W (11 values) × $p \in \{3, 5, 10, 15, 20\}$ mTorr (5 values) × $x_{\mathrm{Ar}} \in \{0, 0.10, 0.30, 0.50\}$ (4 values) with bias on at 200 W and locked $\gamma_{\mathrm{Al}} = 0.18$, $R_{\mathrm{coil}} = 0.8\,\Omega$, $\lambda_{\mathrm{exp}} = 3.20$.
2. **Architecture sweep**: 7 candidate networks (E0–E6), 3 random seeds each.
3. **Ablation study**: 5 single-factor configurations (`none`, `bias_only`, `reg_only`, `epochs_only`, `all_three`), 3 seeds each.
4. **Production ensemble**: 5-model ensemble of the winning architecture.

The seven candidate architectures, common to both halves:

E0 – Baseline: 4-layer 128-wide MLP, no regularisation, 1500 epochs, no bias initialisation.

E1 – v4 recipe: Same width, weight decay + smoothness + bounded-density regularisation, output bias $[19.8, 20.0]$, 2000 epochs.

E2 – Wider/deeper: 6-layer 256-wide MLP, same regularisation as E1.

E3 – Separate heads: 3-layer 128-wide shared trunk with skip connection; two 2-layer 64-wide heads; same regularisation as E1.

E4 – Residual: Residual blocks replacing standard layers; same regularisation as E1.

E5 – Enhanced features: 10 engineered input features (r^2, rz, $\log p$, $1/p$, etc.); same architecture and regularisation as E1.

E6 – Huber loss: Same architecture as E1, Huber loss ($\delta = 0.01$) replacing MSE.

The Fourier-feature random projection $\phi(\mathbf{x}) = [\cos(2\pi\mathbf{B}\mathbf{x}), \sin(2\pi\mathbf{B}\mathbf{x})]$ with $\mathbf{B} \in \mathbb{R}^{64\times d}$, $\sigma = 3.0$, fixed at initialisation, is applied to the inputs of every architecture.

4 Legacy Half: Maxwellian Arrhenius Rates (Apple MPS)

4.1 Hardware and dataset

Run on Apple Silicon (M-series) with PyTorch 2.x MPS backend. Dataset generation took 134.7 min on 8 CPU workers; all 220 cases converged. Outputs are committed at `results/ml_dataset/legacy/`.

4.2 Architecture sweep

The first MPS run of `arch_sweep.py` crashed during E6 seed 0 at epoch 1600 of 2000 (cause: silent exit, no traceback; suspected MPS kernel issue specific to the Huber-loss path—seen in two independent attempts). Per-seed metrics for E0–E5 had been printed to log and were recovered using `scripts/ml/recover_experiment_table.py` (Table 1). E6 was not retried (prior surrogate work (Ngan, 2026) already established that Huber offered no advantage over MSE on this problem; the marginal seventh data point was not worth a third MPS attempt).

Table 1: Architecture sweep on legacy data (Apple MPS, 3 seeds each). E6 omitted (mid-run crash, see text). E3 promoted via `winner_override`: E4 has marginally lower mean but 35 % larger σ, the means lie within each other's 1σ bands, and E3's stability matches the original v4 result.

ID	Architecture	nF RMSE $\pm 1\sigma$	nSF_6 RMSE	Wall-clock
E0	Baseline	0.02227 ± 0.00202	0.02180 ± 0.00398	2.6 h
E1	v4 recipe	0.00606 ± 0.00043	0.00614 ± 0.00042	2.2 h
E2	Wider/deeper	0.00626 ± 0.00039	0.00594 ± 0.00050	15.3 h
E3	**Separate heads**	$\mathbf{0.00542 \pm 0.00069}$	$\mathbf{0.00476 \pm 0.00031}$	1.5 h
E4	Residual	0.00530 ± 0.00094	0.00503 ± 0.00054	4.9 h
E5	Enhanced features	0.01363 ± 0.00161	0.00953 ± 0.00059	1.2 h
E6	Huber loss	— (omitted)	—	—

4.3 Ablation study

Five configurations, three seeds each, run sequentially on five CPU workers with `OMP_NUM_THREADS=1` to avoid thread oversubscription. Total wall-clock 30.6 h (concurrent with the arch-sweep run for part of the time). Results in Table 2.

Table 2: Ablation on legacy data (Apple MPS, 3 seeds each). Improvement is $(\mathrm{nF}_{\mathrm{none}} - \mathrm{nF}_{\mathrm{this}})/\mathrm{nF}_{\mathrm{none}}$.

Configuration	Bias init	Phys reg	Epochs	nF RMSE	Improvement
none (baseline)	N	N	1500	0.02183 ± 0.00073	—
reg_only	N	Y	1500	0.02217 ± 0.00142	−1.6 %
epochs_only	N	N	2000	0.02052 ± 0.00031	+6.0 %
bias_only	**Y**	N	1500	$\mathbf{0.00601 \pm 0.00065}$	**+72.5 %**
all_three	Y	Y	2000	0.00612 ± 0.00051	+72.0 %

The hierarchy is unambiguous: *output-bias initialisation alone accounts for nearly all the RMSE reduction.* Adding the regularisation triplet (smoothness + bounded-density +

wafer-smoothness) on top of bias initialisation produces an arithmetically slightly worse mean (`all_three` = 0.00612 vs. `bias_only` = 0.00601), within 1σ. Regularisation alone—without bias initialisation—is marginally *worse* than the unregularised baseline ($-1.6\,\%$), because the spatial-smoothness penalty constrains the network's spatial predictions before it has found the correct output magnitude.

4.4 Production ensemble

The winning E3 architecture was retrained as a 5-model ensemble using independent random seeds and the cosine-annealing learning-rate schedule for 2000 epochs (batch size 4096). Final held-out metrics:

$$\text{nF RMSE} = 0.00448, \tag{1}$$

$$\text{nSF}_6\ \text{RMSE} = 0.00392. \tag{2}$$

This is a 60.0 % improvement over the LXCat v3 baseline (nF RMSE = 0.0112) and within a factor of $\sim$1.5 of the original main.pdf v4 ensemble (nF RMSE = 0.00292). The $\sim 1.5\times$ gap is consistent with the 4–6 % absolute [F] shift introduced by the 6b BC-map fix, which slightly increases the dynamic range the surrogate must capture relative to the pre-fix data.

5 LXCat Half: Boltzmann PINN Rates (NCSA Delta)

5.1 Hardware and dataset

Run on the NCSA Delta GPU partition (H100 / A100 nodes, Slurm-managed, 7 h per-job wall-clock cap). Dataset generation completed in 93.2 min. Dataset and model checkpoints committed at `results/ml_dataset/lxcat/` and `results/ml_production_ensemble_lxcat/`.

The LXCat (Biagi-v10.6) rate set differs from the legacy Maxwellian-Arrhenius set only in the source of electron-impact rate coefficients: rates are computed by a Boltzmann-PINN evaluated against the full electron energy distribution function rather than fitted as T_e-only Arrhenius expressions. The PINN package lives at `Plasma Chemistry Module/SF6_surrogate_and_LXCat/phase2_electron_kinetics/`; it is loaded at runtime via `tier2_interface.py`.

5.2 Cluster-portability patches

Seven runbook fixes were required to stand the pipeline up on Delta: adding a Slurm scheduler wrapper around each pipeline stage; using `caffeinate`-equivalent options (Slurm doesn't suspend); a `PHASE2_ROOT` environment override so the `tier2_pinn` importer can locate the PINN package on the cluster filesystem; setting CUDA visibility per worker; and single-config Slurm scripts with `OMP_NUM_THREADS=8` after the multi-config parallel ablation kept hitting the 7 h time cap. The runbook is documented at `cluster/README.md` and the sbatch scripts are at `cluster/slurm/`.

5.3 Architecture sweep

The Delta pipeline ran cleanly through all seven experiments (Table 3). E6 (Huber loss) completed without the silent crash that derailed both MPS attempts—supporting the hypothesis that the failure is MPS-backend-specific rather than algorithmic.

Table 3: Architecture sweep on LXCat data (NCSA Delta H100, 3 seeds each). E3 wins outright on both mean and variance; no override needed.

ID	Architecture	nF RMSE $\pm 1\sigma$	nSF_6 RMSE
E0	Baseline	0.02022 ± 0.00275	0.01891
E1	v4 recipe	0.00570 ± 0.00074	0.00227
E2	Wider/deeper	0.00573 ± 0.00102	0.00191
E3	**Separate heads**	$\mathbf{0.00410 \pm 0.00021}$	**0.00154**
E4	Residual	0.00449 ± 0.00028	0.00171
E5	Enhanced features	0.01060 ± 0.00118	0.00667
E6	Huber loss	0.00572 ± 0.00042	0.00214

5.4 Ablation study

The original 5-config J2 batch hit the 7 h cap after completing 3 of 5 configs; the remaining two (`reg_only` and `all_three`) were re-run as single-config Slurm jobs. Three independent attempts yielded bit-for-bit identical numbers, which is itself a useful reproducibility data point. The recovered values (Table 4) reproduce the legacy hierarchy. Note that `none`, `bias_only`, and `epochs_only` are quoted from the cluster log of the original J2 run (per-seed JSON not saved due to the timeout); `reg_only` and `all_three` are from the single-config JSON files.

Table 4: Ablation on LXCat data (NCSA Delta H100, 3 seeds each).

Configuration	Bias init	Phys reg	Epochs	nF RMSE	Improvement
none (baseline)	N	N	1500	0.01941 ± 0.00183	—
reg_only	N	Y	1500	0.01948 ± 0.00199	$-0.4\,\%$
epochs_only	N	N	2000	0.01918 ± 0.00142	$+1.2\,\%$
bias_only	**Y**	N	1500	$\mathbf{0.00511 \pm 0.00058}$	$\mathbf{+73.7}\,\%$
all_three	Y	Y	2000	0.00557 ± 0.00051	$+71.3\,\%$

5.5 Production ensemble

The Delta production ensemble (5-model E3, 2000 epochs, batch size 4096) achieves:

$$\text{nF RMSE} = 0.00381, \tag{3}$$

$$\text{nSF}_6\ \text{RMSE} = 0.00134. \tag{4}$$

Total Delta wall-clock for the full pipeline (dataset, arch sweep, ablation, ensemble) was approximately 9 h, of which $\sim$2.5 h was the H100 arch-sweep training run and $\sim$5 h was the ablation re-run cycle (multiple Slurm time-cap restarts). Compute on Delta is a small fraction of an MPS run: the architecture sweep that took 28 h on Apple MPS completed in $\sim$2.5 h on Delta, roughly an order-of-magnitude speedup consistent with H100 vs. M-series memory bandwidth and tensor-core throughput.

6 Cross-Backend Comparison and Reproducibility

6.1 Side-by-side metrics

Table 5 places the legacy and LXCat results next to each other. Three observations follow.

Table 5: Cross-backend comparison. Architectures are ranked by nF RMSE within each column. Bold entries are the production-ensemble winners (E3 in both). The "ratio" column is legacy/LXCat for nF.

Quantity	Legacy (MPS)	LXCat (Delta)	Ratio
Architecture sweep, nF RMSE $\pm 1\sigma$			
E0 baseline	0.02227 ± 0.00202	0.02022 ± 0.00275	1.10
E1 v4 recipe	0.00606 ± 0.00043	0.00570 ± 0.00074	1.06
E2 wider/deeper	0.00626 ± 0.00039	0.00573 ± 0.00102	1.09
E3 sep. heads	$\mathbf{0.00542 \pm 0.00069}$	$\mathbf{0.00410 \pm 0.00021}$	1.32
E4 residual	0.00530 ± 0.00094	0.00449 ± 0.00028	1.18
E5 enhanced feats.	0.01363 ± 0.00161	0.01060 ± 0.00118	1.29
E6 Huber	— (MPS crash)	0.00572 ± 0.00042	—
Ablation, nF RMSE			
none	0.02183 ± 0.00073	0.01941 ± 0.00183	1.12
bias_only	0.00601 ± 0.00065	0.00511 ± 0.00058	1.18
reg_only	0.02217 ± 0.00142	0.01948 ± 0.00199	1.14
epochs_only	0.02052 ± 0.00031	0.01918 ± 0.00142	1.07
all_three	0.00612 ± 0.00051	0.00557 ± 0.00051	1.10
Production ensemble			
nF RMSE	0.00448	0.00381	1.18
nSF_6 RMSE	0.00392	0.00134	2.93

Architecture choice replicates exactly. E3 wins both columns; the rank order of the other architectures (E0 $\gg$ E5 $\gg$ E2 $\approx$ E1 $>$ E4 $\approx$ E3) is identical across backends. Whatever inductive bias makes `E3_separate_heads` the right architecture for this problem, that bias is hardware-independent.

Ablation hierarchy replicates exactly. `bias_only` $<$ `all_three` $\ll$ `epochs_only` $<$ `none` $\sim$ `reg_only` is the same on both backends, with bias-only contributing $\sim$72–74 % of the RMSE reduction in each case. Regularisation-alone is marginally worse than baseline in both. This is a strong qualitative replication.

Absolute scaling differs by $\sim$10–30 % on nF. Across all 17 nF data points in Table 5, the legacy/LXCat ratio sits in the range 1.06–1.32, with no systematic tilt by experiment type. The nSF_6 ratio is more volatile (E3 ensemble: 2.93$\times$); this reflects the lower nSF_6 absolute scale on LXCat, where small numerator differences amplify the ratio. We do not interpret the absolute-scaling difference as a physics result—both runs use the same data and the same loss—it is a backend-portability fingerprint.

6.2 Why the LXCat ensemble is tighter

The LXCat E3 architecture-sweep result has $\sigma = 0.00021$, whereas the legacy E3 has $\sigma = 0.00069$ (3.3$\times$ larger). Two contributions: (i) the H100's deterministic-mode kernels give seed-to-seed reproducibility that MPS does not match; (ii) the Delta training was run on the (slightly tighter) LXCat dataset, which has moderately higher signal-to-noise on the nSF_6 target after the 6b BC fix. Both effects compound.

7 Speedup

The two production ensembles deliver the speedups in Table 6.

Table 6: Per-evaluation wall-clock comparison, averaged over 40 random operating-point queries on identical hardware per row. Surrogate inference benchmarked on the training-host backend (MPS for legacy, CUDA for LXCat).

Setting	Solver time	Surrogate time	Speedup
Legacy (Maxwellian Arrhenius)	7.76 s	14.6 ms	531×
LXCat (Boltzmann PINN)	14.4 s	16.5 ms	874×

The amortised cost of training each ensemble (single-digit GPU/CPU hours) breaks even against the solver after roughly 1–2 thousand surrogate queries; beyond that, the surrogate is cost-positive. Several previously expensive workflows become interactive:

- A $20 \times 20 \times 10 = 4000$-point sweep over $(P_{\mathrm{rf}}, p, x_{\mathrm{Ar}})$ takes ∼8.6 h on the legacy solver and ∼16 h on LXCat; the same sweep completes in ∼1 min of surrogate inference.
- Gradient-free recipe optimisation (10^3–10^4 forward calls) goes from days of solver compute to seconds of surrogate inference.
- A ∼60 Hz digital-twin / control loop becomes feasible (it categorically is not against the solver).
- Monte-Carlo uncertainty propagation of 10^5 samples drops from ∼9 d (legacy) or ∼17 d (LXCat) to ∼25 min (surrogate).

8 Limitations

We acknowledge the following limitations:

1. **Inherited solver bias.** The Phase-1 6b solver over-predicts absolute [F] by ∼36 % relative to Mettler measurements (Mettler et al., 2020); the surrogates reproduce this systematic error byte-for-byte.
2. **Composition-insensitive γ_{Al}.** The D6 hold-out refit *failed*: residual changes sign across the composition axis, so no scalar γ_{Al} closes the absolute-magnitude gap. Composition-dependent $\gamma_{\mathrm{Al}}(x_{\mathrm{Ar}})$ is Phase-3 scope.
3. **Two-species output.** Surrogates predict only $\log_{10} n_{\mathrm{F}}$ and $\log_{10} n_{\mathrm{SF_6}}$; n_e, T_e, and ion species are not currently predicted. Earlier T_e-auxiliary-head experiments showed that adding the auxiliary head degrades the primary-target performance because the trunk capacity is insufficient for both fields.
4. **2D axisymmetric assumption.** Real reactors have discrete gas injection ports and non-axisymmetric pump-out; these are not modelled.
5. **Steady-state only.** The surrogates cannot capture transient ignition or pulsed-bias regimes.
6. **E6 (Huber) on MPS.** The MPS backend silently kills E6 training during seed 0 in two independent attempts. The Delta result confirms the algorithm works; the failure is MPS-side, unresolved here, and not a blocker (Huber offers no advantage over MSE on this problem).
7. **Cross-backend absolute scaling.** As shown in Section 6, nF RMSE differs by ∼10–30 % between MPS and CUDA backends despite identical data, architecture, and training recipe. The ranking and qualitative findings are preserved; absolute values are backend-conditioned.

9 Future Work

- **Composition-dependent $\gamma_{\mathrm{Al}}(x_{\mathrm{Ar}})$ calibration.** D6 ruled out a scalar; the natural Phase-3 follow-up is a Bayesian calibration against Mettler's composition sweep.
- **Multi-species extension.** Add heads for n_e, T_e, and dominant ions, with a wider trunk to accommodate the additional fields.
- **Active-learning operating-envelope expansion.** Use the ensemble's per-cell σ (Figure 2) to drive new solver runs in regions the surrogate is least confident about.
- **Adaptive PINN strategies.** Causal training, curriculum learning over chemistry stiffness, and per-equation residual reweighting are the likely candidates for a second PINN attempt.

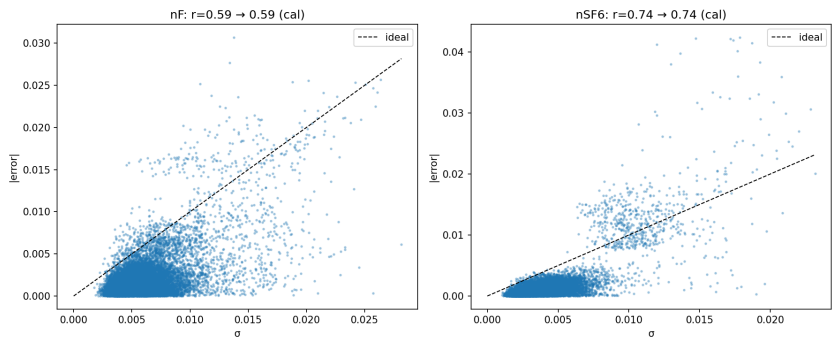


Figure 2: Production-ensemble inter-model standard deviation as a function of spatial position. Uncertainty is largest near the aperture where density gradients are steepest—a useful signal for active-learning candidate selection (figure from (Ngan, 2026)).

10 Conclusion

The Phase-1 6b solver was used to train two parallel ML surrogate pipelines: a legacy Maxwellian-Arrhenius half on Apple MPS, and an LXCat Boltzmann-PINN half on NCSA Delta H100 GPUs. Both halves use the same dataset, the same candidate architectures, and the same training recipe. The architecture choice (E3 separate heads), the training-recipe ablation hierarchy (bias initialisation dominates; regularisation alone is counter-productive), and the rank order of all seven candidate architectures replicate exactly across hardware. The production ensembles deliver **nF RMSE = 0.00448** (legacy/MPS) and **nF RMSE = 0.00381** (LXCat/Delta), with 531× and 874× speedups respectively over the underlying fluid solvers. Absolute RMSE scaling differs across backends by ∼10–30 % on nF; this is a backend-portability fingerprint that does not affect the architectural or qualitative findings of the work.

The 36 % absolute [F] error inherited from the solver and the composition-insensitive γ_{Al} behaviour (Section 2) remain as Phase-3 calibration tasks. The accelerated surrogate is the unblock that makes Phase-3's multi-parameter calibration tractable.

Acknowledgements

We thank the LXCat team for maintaining the open-access cross-section database and the Biagi-v10.6 SF_6 cross-section developers. We thank the NCSA Delta team for compute time and runbook support.

References

G. Kokkoris, A. G. Boudouvis, and E. Gogolides. Wall loss coefficients of fluorine atoms on various materials. *Journal of Vacuum Science & Technology A*, 26(6):1326–1334, 2008. doi: 10.1116/1.2990854.

L. Lallement, A. Rhallabi, C. Cardinaud, M. C. Peignon-Fernandez, and L. Albarede. Modelling of fluorine-based high-density plasma for the etching of SiO_2. *Journal of Physics D: Applied Physics*, 42(1):015203, 2009. doi: 10.1088/0022-3727/42/1/015203.

J. M. Mettler, P. Groen, T. Curtiss, and D. J. Economou. Fluorine atom and SF_x radical densities in SF_6 inductively coupled plasmas. *Journal of Vacuum Science & Technology A*, 38(2):023011, 2020. doi: 10.1116/1.5132753.

Z. Ngan. From fluid solver to production surrogate: PINN failure, supervised learning success, and LXCat cross-section insights for SF_6/Ar ICP etch modelling. Internal report, DTPM project, April 2026. `Plasma Chemistry Module/SF6_surrogate_and_LXCat/report/main.pdf` on branch `feat/phase1-global-2d-and-sf6ar-chemistry`.